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The Chinese University of Hong Kong
Department of Information Engineering

VLMaterial: Vision-Language Model-Based Camera-Radar Fusion for Physics-Grounded Material Identification

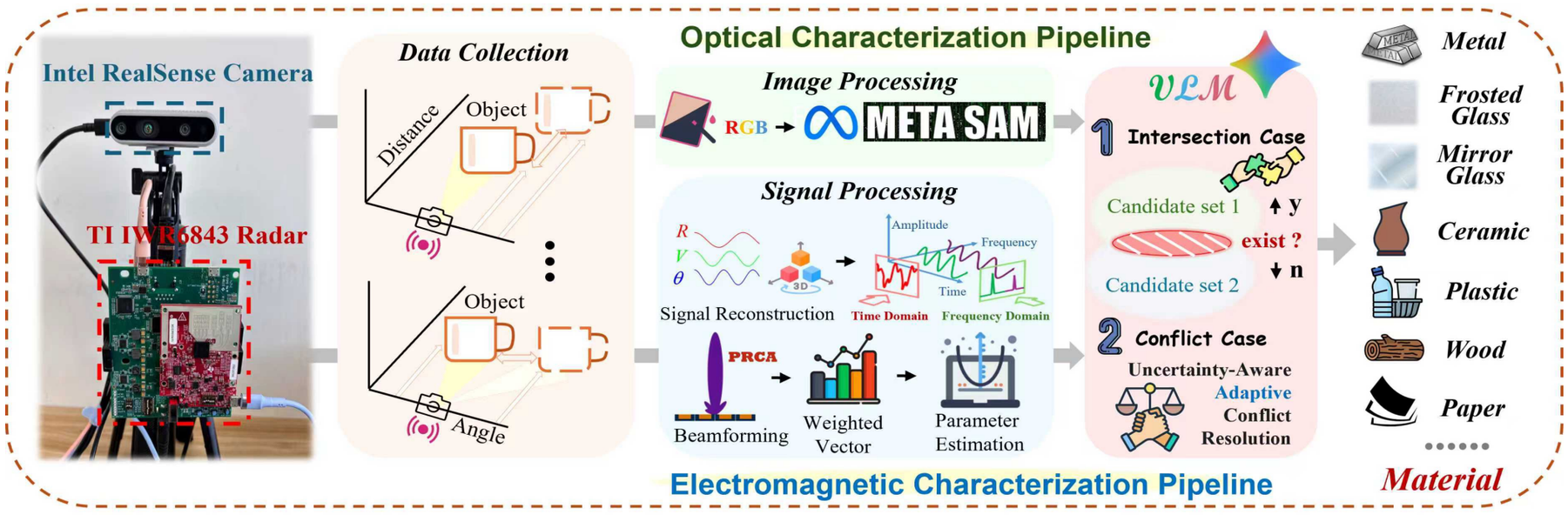
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Accurate **material recognition** is critical for intelligent systems, e.g., visual similarity may cause a robot to mistake a plastic cup for glass and microwave it unsafely.
While mmWave radar offers robust sensing regardless of lighting, existing camera-radar fusion **lacks semantic interpretability** and is **limited to closed-set** categories.
We propose **VLMaterial**, a training-free framework that uses VLM to fuse camera and radar data for **physics-grounded** material identification.

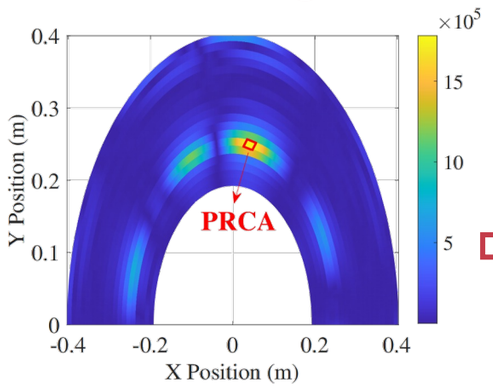
Framework of VLMaterial



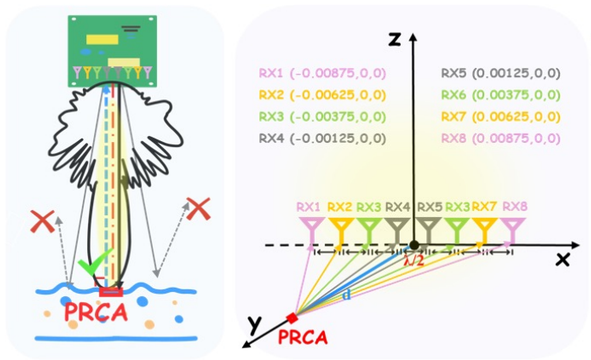
Main contributions:

- ***Electromagnetic Characterization.*** PRCA, beamforming, and weighted vector synthesis extract **dielectric constant** from raw radar signals.
- ***Adaptive Fusion Mechanism.*** Balances camera and radar via intersection consensus and uncertainty-aware conflict resolution.
- ***SOTA Performance.*** VLMaterial achieves **96.08% accuracy** on 41 objects + 4 counterfeits without training, outperforming CRFUSION, LLMaterial, and unimodal baselines.

Electromagnetic characterization pipeline

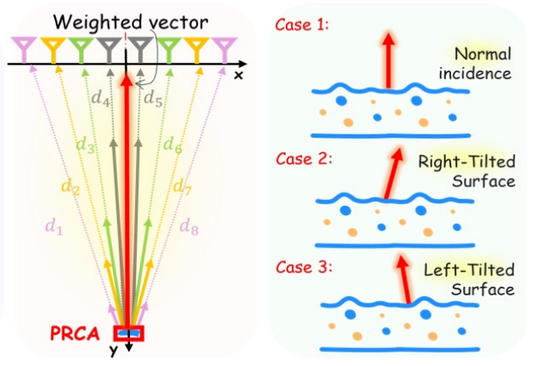


$$SNR = \frac{P_r}{P_n} = \frac{P_t G_t G_r \lambda^2 \sigma}{(4\pi)^3 k T_n B R^4}$$

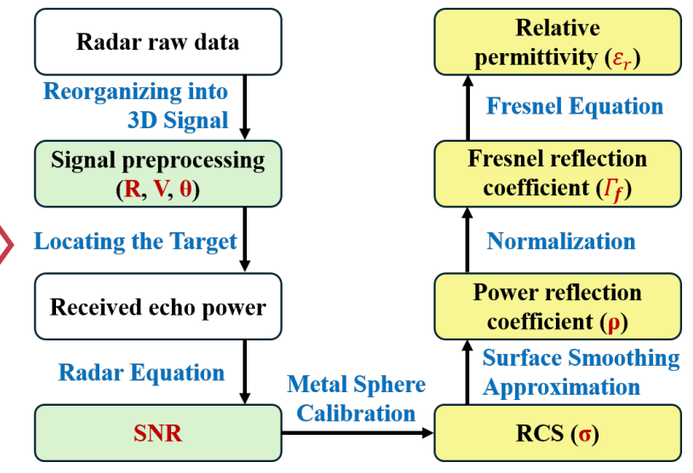


(a) Specular reflection model. (b) Antenna array geometry.

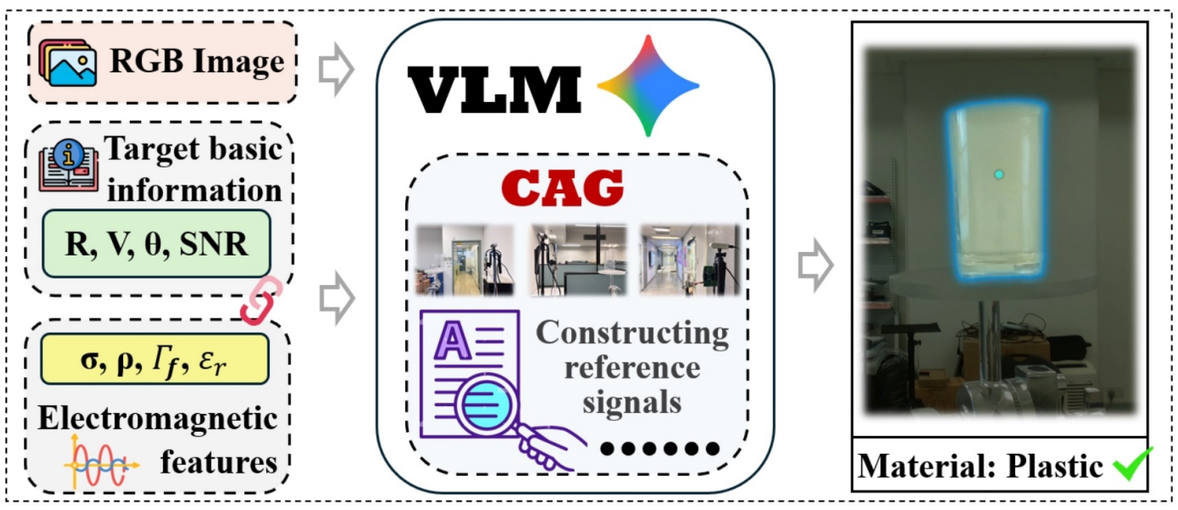
Weighted vector synthesis



(c) Weighted vector summation. (d) Surface orientation detection.



Context-Augmented Generation (CAG)



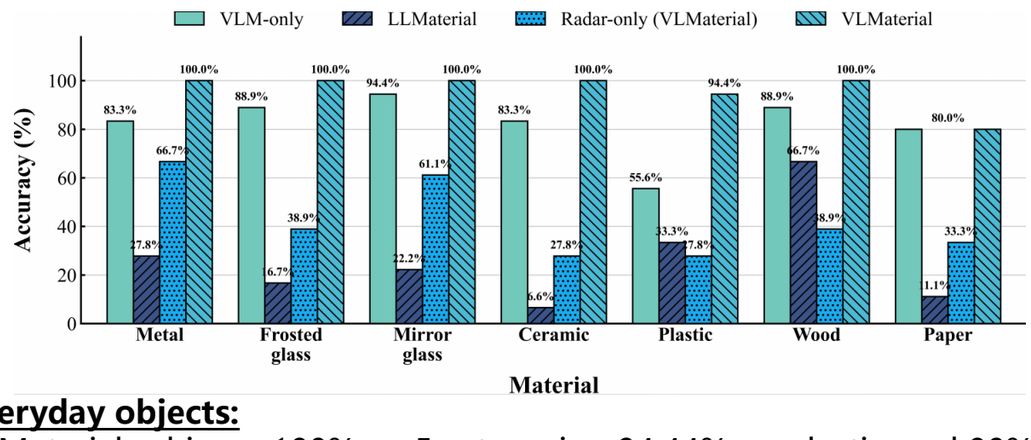
Adaptive fusion mechanism

Intersection vs. Conflict:
Intersection leads to direct consensus; disjoint sets trigger uncertainty-aware adaptive conflict resolution using real-time confidence scores.

Visual Uncertainty:
Low luminance, clutter, and VLM hesitation are weighted for confidence.

Radar Uncertainty:
Low SNR, distance, and oblique angle are weighted for confidence.

Performance Evaluation



Everyday objects:
VLMaterial achieves 100% on 5 categories, 94.44% on plastic, and 80% on paper. VLM-only (55.56%–94.44%) and radar-only (27.78%–66.67%) perform worse. VLMaterial significantly outperforms all baselines.



Counterfeit objects:
VLM-only fails entirely on all counterfeits, radar-only suffers confusion (misidentify plastic case 4 as wood), VLMaterial correctly identifies all counterfeits.